

Generation Interconnection System Impact Study Report

For Queue Project Q23-2313

Arcogen 138 kV

May 02, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q23-2313. The Interconnection Customer proposes a 305.7 MW solar photovoltaic facility interconnecting at the Arcogen 138 kV substation in the Virginia area. The request was received on May 09, 2023. The study evaluated system intact and single contingency conditions using the regional planning case, including 682 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 7 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$25,495,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Arcogen - Chincoteague 138.0 kV	Southtown - Tasley 345.0 kV line	159.2	386	89.4
Longmire - Orange Bulk Substation 138.0 kV	Cady - Easton 2 345.0 kV	115.1	196	5.3
Belle Haven Diesel Generation Facility - Chincoteague 138.0 kV	Southtown - Commonwealth Chesapeake 345.0 kV	114.4	199	11.3
Menomonie Substation - Wattsville 138.0 kV	Tasley 345 kV/138 kV XFMR	111.2	570	100.0
Chincoteague - Wattsville 138.0 kV L	Tasley 345 kV/138 kV Transformer Bank	108.1	587	100.0
Orange Bulk - Donner 138.0 kV	Cady - Dodd Road 345.0 kV	105.8	291	7.4
Commonwealth Chesapeake - Menomonie 138.0 kV	Tasley 345-138 kV XFMR	103.6	590	100.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Arcogen - Chincoteague 138.0 kV	Rebuild line with higher-capacity conductor	\$9,797,000	63% (\$6,194,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Longmire - Orange Bulk Substation 138.0 kV	Rebuild line with higher-capacity conductor	\$8,318,000	8% (\$626,000)
Belle Haven Diesel Generation Facility - Chincoteague 138.0 kV	Rebuild line with higher-capacity conductor	\$39,150,000	20% (\$7,927,000)
Menomonie Substation - Wattsville 138.0 kV	Rebuild line with higher-capacity conductor	\$10,165,000	60% (\$6,124,000)
Chincoteague - Wattsville 138.0 kV L	Reconductor line	\$3,029,000	60% (\$1,825,000)
Orange Bulk - Donner 138.0 kV	Reconductor line	\$10,384,000	7% (\$688,000)
Commonwealth Chesapeake - Menomonie 138.0 kV	Reconductor line	\$3,504,000	60% (\$2,111,000)
	Total allocated to project		\$25,495,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.